

Course Specification

Name of Institution
Faculty/Program
Innovation

National Institute of Development Administration
Graduate School of Communication Arts and Management

Section 1 General Information

- 1. Course Code and Title** **CM 7005 Reality Technologies and Rapid User Experience Design for Communication**
- 2. Total Credits** 45 Credit Hours
- 3. Program and Course Type**
Major Course
- 4. Course Instructor**
4.1 Instructor Name Associate Professor Dr. Chutisant Kerdvibulvech
- 5. Semester / Year of Study**
Semester 1 / Year of study 2021
- 6. Pre-requisite Course (if any)**
NA
- 7. Co-requisites Course (if any)**
NA
- 8. Venue**
Nawamin Building, National Institute of Development Administration
- 9. Latest Date of Course Specification Development or Modification**
1 August 2021

Section 2 Aims and Objectives

1. Course Aims

This course is aimed to enhance students' ability to apply theories and concepts to practice and research in communication and innovation.

2. Courses Objectives

- 1.To provide students theories and concepts of communication and innovation.
2. Explain the role of the communication theories in the communication and innovation environments.
3. Increase students' ability to apply communication and innovation theories and concepts to communication and innovation research.

3. Reason for Course Development/Modification

The course content was changed by following Thailand 4.0 policy. Several theories and framework related to information communication and technology was added.

Section 3 Course Description and Implementation

1. Course Description

Understanding different types of digital media such as new digital media, website and search engine optimization, artificial intelligence, augmented reality and virtual reality. Analyzing the current trends for digital media production for management innovation. Creating several specific types of aforementioned digital media using modern information technology. This course will integrate a comprehensive introduction between reality technologies (such as augmented reality, virtual reality, diminished reality, etc.) and rapid user experience design, focusing on communication. The combination knowledges are beneficial in facilitating any recent communication project in the digital age to shape the dynamic world of new communications.

2. Number of Hour per Semester

Number of Lecture Hours: 3 hours per week

3. Number of hours provided for academic advice and guidance to individual student

3 hours per week will be communicated through Facebook messenger and face to face interaction

Section 4 Development of Student Learning Outcome

1. Students will understand the communication theories and concepts at all levels including intrapersonal communication, interpersonal communication, organizational communication, intercultural communication, mass communication, persuasive communication and information and communication technology.

2. Students will be able to apply the communication and emerging technology theories and concepts into professional communication and technology practices.

3. Students will be able to use communication theories and model to guide the design of the communication and emerging technology researches.

1. Morals and Ethics

1.1 Morals and Ethics to be developed

- Media ethics and literacy

1.2 Teaching Methods

- Lecture
- Discussion

1.3 Evaluation Methods

- Observation

2. Knowledge

2.1 Knowledge to be acquired

- Communication and emerging technology theories

2.2 Teaching Methods

- Lecture
- Group Discussion

- 2.3 Evaluation Methods
 - Group Assignment

3. Intellectual Development

- 3.1 Intellectual Development to be developed
 - Application of theory to practice and research
- 3.2 Teaching Methods
 - Group assignment
- 3.3 Evaluation Methods
 - Rubrics

4. Interpersonal Skills and Responsibilities

- 4.1 Interpersonal Skills and Responsibilities to be developed
 - NA
- 4.2 Teaching Methods
 - NA
- 4.3 Evaluation Methods
 - NA

5. Numerical Analysis, Communication, and Information Technology Skills

- 5.1 Numerical Analysis, Communication, and Information Technology Skills to be developed
 - NA
- 5.2 Teaching Methods
 - NA
- 5.3 Evaluation Methods
 - NA

6. Mapping

Code	Courses	Credit	Ethics	Knowledge	Cognitive		Interpersonal	TQF 3 Analytical skills
			ELO 6	ELO 1	ELO 2	ELO 5	ELO 4	ELO 3
7005	Reality Technologies and Rapid User Experience Design for Communication	3		E	I	P		

I = introductory, E = Emphasis, A=Assessment, P=Partial relevant

Alignment between program ELOs, teaching method, assessment method and achievement indicator

Program ELOs	Teaching methods	Assessment methods	Achievement indicators
ELO 1: Apply innovative communication theories and concepts into professional communication and technological practices in digital environments.	-Lecture -Group discussion -Written assignment	-Assessment rubrics -Written examination	-Ability to explain communication and innovation theory and concept -Ability to apply theory and concept to the practice
ELO 2: Formulate research in the areas of communication and management innovation.	-Written assignment -Oral presentation	-Assessment rubrics	-Ability to apply theory and concepts to the communication research.
ELO 3: Create content and communication design that responses to the communication industrial trends	NA	NA	NA
ELO 4: Develop social digital network for communication and management innovation.	NA	NA	NA
ELO 5: Propose innovative solutions in professional communication and digital practice.	NA	NA	NA
ELO 6: Choose international and national standardize values and professional codes of ethic and the nobility of profession in communication and digital practices.	NA	NA	NA

Section 5 Teaching Plan and Evaluation Plan

1. Teaching Plan*

W	Regular Program (13.00-16.00)	Special Program	Lecture Topic	H	Teaching activities	Instructor
1	20 Aug 21	-	Introduction to Digital Media Production	3	Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech
2	27 Aug 21	-	History, Creativity and Research for Digital Media Production	3	Lecture Group presentation / E-learning (Microsoft Teams)	Assoc. Prof. Dr. Chutisant Kerdvibulvech
3	3 Sep 21	-	Recent Applications for Digital Media Production	3	Lecture Group Discussion / E-learning (Microsoft Teams)	Assoc. Prof. Dr. Chutisant Kerdvibulvech
4.	10 Sep 21	-	Recent Applications for Digital Media Production (cont.)	3	Lecture Group Discussion Research based learning / E-learning (Microsoft Teams)	Assoc. Prof. Dr. Chutisant Kerdvibulvech
5	17 Sep 21	-	Recent Applications for Digital Media Production (cont.)	3	Lecture Group Discussion Research based learning / E-learning (Microsoft Teams)	Assoc. Prof. Dr. Chutisant Kerdvibulvech
6	24 Sep 21	-	Current Trends for Digital Media Production	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech
7.	1 Oct 21	-	Introduction to New Digital Media	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech
8.	15 Oct 21	-	Recent New Digital Media	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech / Guest Speaker

W	Regular Program (13.00-16.00)	Special Program	Lecture Topic	H	Teaching activities	Instructor
			Midterm Examination Week			

1. Teaching Plan (Continued)

W	Regular Program	Special Program	Lecture Topic	H	Teaching activities	Instructor
9	22 Oct 21	-	Recent New Digital Media (cont.)	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech / Guest Speaker
10	29 Oct 21	-	Recent New Digital Media (cont.)	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech / Guest Speaker
11	5 Nov 21	-	User Experience / Website Creation for Digital Media	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech
12	12 Nov 21	-	User Experience / Website Creation for Digital Media (cont.)	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech
13	19 Nov 21	-	User Experience / Search Engine Optimization for Digital Media Production	3	Lecture Group Discussion	Assoc. Prof. Dr. Chutisant Kerdvibulvech
14	TBA	-	Recent Technologies for Communication (Special Lecture)	3	Lecture Group Discussion Research based learning / E-learning (Microsoft Teams)	Assoc. Prof. Dr. Chutisant Kerdvibulvech
15	3 Dec 21	-	User Experience / Search Engine Optimization for Digital Media Production (cont.)	3	Group presentation	Assoc. Prof. Dr. Chutisant Kerdvibulvech

* Note that in some weeks, the lecture topic may be changed.

2. Evaluation plan

Activities	Learning Outcome	Evaluation Methods	Week of Evaluation	Mark Allocation
1	1,2, 6	Class participation and	1-15	10 %

		attendance		
2	1	Group project	1-5	30 %
3	1	Midterm project	6-10	30 %
4	1	Final project	11-15	30 %

3. Attendance and Participation

Students are required to attend a minimum of 80% of all sessions and expected to participate in group discussions and activities. Students are also expected to arrive at the class no later than 15 minutes after class has started.

Section 6 Teaching and Learning Resource

1. Reading materials

Required Text	- New Digital Media..Media for the Future, Author: Chutisant Kerdvibulvech, 2nd Edition (2017), NIDA Publishing - Augmented Reality: A Modern Vision-Based Approach, Author: Chutisant Kerdvibulvech, 2nd Edition (2016) - Mixed Reality: Merging Real and Virtual Worlds, Yuichi Ohta, Hideyuki Tamura, 418 pages, Springer, ISBN-13: 978-3642875144 (April 15, 2014)
Additional Reading	- C. Kerdvibulvech, "An Innovative Real-time Mobile Augmented Reality Application in Arts," The 4th International Conference on Augmented Reality, Virtual Reality and Computer Graphics (SALENTO AVR'17), Lecture Notes in Computer Science (LNCS), Part II, vol. 10325 (DBLP, SCImago Q2 and Scopus), Springer International Publishing AG, Ugento (Lecce), Italy, pp. 251-260, June 12-15, 2017. Oral - C. Kerdvibulvech, "Vision and Virtual-based Human Computer Interaction Applications for a New Digital Media Visualization", 23rd International Conference on Computer Graphics, Visualization and Computer Vision 2015 (WSCG), Eurographics Association & ACM SIGGRAPH (Scopus, SCImago and Thomson Reuters ISI), Plzen, Czech Republic, June 8 - 12, 2015. - C. Kerdvibulvech, "An Innovative Use of Multidisciplinary Applications Between Information Technology and Socially Digital Media for Connecting People", The Multidisciplinary International Social Networks Conference 2015 (Springer), in Communication of Computer and Information Science (CCIS), (Scopus and SCImago Quartile 4th), Ehime, Japan, September 1 – 3, 2015.

Research Journal	C. Kerdvibulvech, "Hand Tracking by Extending Distance Transform and Hand Model in Real-time," Pattern Recognition and Image Analysis, Vol. 25, Issue 3, pp. 437-441, Springer Publisher (INSPEC, SCImago Q3, Elsevier's Scopus, EiCompendex, ACM, and Google Scholar), July 2015.
Electronic website	http://www.ieee.org

2 .Research publications

Research publication	Week
-	-

3 .Publication, body of knowledge and academic services that integrated in the teaching activities

Publication, body of knowledge and academic service	Week
-	-

Section 7 Course Evaluation and Improvement

1. Evaluation Strategies for Course Effectiveness by Students

Student comments and suggestions are applied in evaluation strategies for course effectiveness.

2. Teaching Evaluation Strategies

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3. Teaching Improvement

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4. Verification of Students Achievements in the Course

This group assignment is aimed to help students get started with the communication and technology theories. Student will learn how to apply communication and technology theories to develop possible applications.

Individual & Group Assignment Paper Rubric

The 6 Aspects for Assessment for 3 projects (30%*3 = 90%)

1. Relevance to the course	0 1 2 3 4 5
2. Effectiveness of the used concepts and theories	0 1 2 3 4 5

3. Additional literature sources and research	0 1 2 3 4 5
4. Reliability literature sources	0 1 2 3 4 5
5. Originality and creativity	0 1 2 3 4 5
6. Communication and clarity presentation	0 1 2 3 4 5

The other 10% is for class participation and attendance.

5. Course Review and Improvement Plan

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6. Contact

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